

$$(ax^2 + bx + c) \sim \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

$$(ax^2 + bx + c)' = 2ax + b$$

$$(2ax + b)' = 2a$$

$$\varphi \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 2a \\ 2a + b \\ 2b \end{pmatrix}$$

A ... matice operátoru φ

$$A = \begin{pmatrix} 2 & 0 & 0 \\ 4 & 0 & 0 \\ 0 & 2 & 0 \end{pmatrix} \quad \text{char. polynom: } -\lambda^3 + 2\lambda^2 = \lambda^2(2 - \lambda)$$

vlastní čísla: 0, 2

$$\text{Ker } A = \text{Ker} \begin{pmatrix} 2 & 0 & 0 \\ 4 & 0 & 0 \\ 0 & 2 & 0 \end{pmatrix} = \text{span} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$J = \begin{pmatrix} \boxed{2} & 0 & 0 \\ 0 & \boxed{0} & 1 \\ 0 & 0 & \boxed{0} \end{pmatrix}$$

geom násobnost 0 1

$$B = \left(\begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix} \right) \sim (x^2 + 2x + 2, 1, \frac{x}{2})$$